WEEK 1 OF 15 · STAGE 1: FOUNDATIONS

Why model the future?

What an integrated assessment model is and who uses it

New R skill Install R and RStudio; console, scripts, projects, packages

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 — Foundations

- **Week 1 — Why model the future?** (current)
- Week 2 — Climate primer
- Week 3 — Energy primer
- Week 4 — Economics primer
- Week 5 — Cost-benefit IAMs: DICE
- Week 6 — The scenario framework

Stage 2 — GCAM in depth

- Week 7 — Inside GCAM
- Week 8 — Energy in GCAM
- Week 9 — Land, water and bioenergy
- Week 10 — Emissions and policy

Stage 3 — Advanced methods

- Week 11 — Optimisation IAMs
- Week 12 — Why models disagree
- Week 13 — Uncertainty
- Week 14 — Frontiers and limits
- Week 15 — Projects and review

TODAY'S GOALS

By the end of today you can

1. Explain in plain words what an integrated assessment model (IAM) does
2. Name the main inputs and outputs of an IAM
3. Tell the two model families apart: cost-benefit and process-based
4. Install R and RStudio and run a first script

? **Big question** Climate reports show charts running to 2100. How can anyone say anything about 2100 when we cannot forecast the weather two weeks ahead?

Who can see 2100?

A weather forecast is unreliable two weeks ahead. Yet climate reports are full of charts that run to 2100.



If nobody can predict the future, what are those charts actually telling us?

AN EVERYDAY PICTURE

An IAM is like a recipe calculator



You type in the ingredients: how many people, how rich they are, how cheap solar is, which policies exist. The calculator works out a cake that adds up: energy used matches energy made, emissions match fuel burned, warming matches emissions.

What it means for models

1. Inputs are assumptions you choose
2. The model keeps every number consistent
3. Change an ingredient and the cake changes

HOW IT WORKS

What goes in and what comes out

1

Inputs

Population, income, technology costs, resources, policies

2

The model

Matches supply and demand, counts emissions

3

Climate

Turns emissions into warming

4

Outputs

Energy mix, emissions, land, prices, temperature

Policies are an input. The model never decides what governments will do; you tell it.

KEY POINTS

Why the word 'integrated'?

- Energy, economy, land, water and climate sit in one model
- A change in one system shows up in the others
- Example: more bioenergy means more cropland, which can mean less forest
- 'Assessment' means the purpose is to inform decisions
- So an IAM is a tool for asking what-if questions about this century

Worked example: why consistency matters

Problem The world uses 600 units of energy a year. Each unit releases 0.06 Gt of CO2. A policy cleans up the energy mix so each unit releases 0.03 Gt. How much do emissions fall?

1. Before: 600 × 0.06 = 36 Gt of CO2
2. After: 600 × 0.03 = 18 Gt of CO2
3. Change: 36 − 18 = 18 Gt
4. As a share: 18 ÷ 36 = 0.5, so a 50% cut

Answer

Emissions fall by 18 Gt, a 50% cut (illustrative numbers)

SIDE BY SIDE

Two families answer different questions

	Cost-benefit models	Process-based models
Question	How much should we cut?	How can a goal be met?
Method	Weigh cutting cost vs damage	Detailed energy and land sectors
Size	A few equations, one world	Thousands of inputs, many regions
Examples	DICE, FUND, PAGE	GCAM, MESSAGEix, REMIND, IMAGE
In this course	Week 5	Weeks 7 to 13

DICE fits on a page; GCAM has thousands of inputs. Different tools for different questions.

Quick check

Which question is a job for a process-based model like GCAM?

A What is one more tonne of CO2 worth in damage?

B Can Japan reach net zero without new nuclear?

C Will it rain in Trondheim next Tuesday?

D What should the global discount rate be?

Quick check: answer

Which question is a job for a process-based model like GCAM?

A. What is one more tonne of CO2 worth in damage?

B. Can Japan reach net zero without new nuclear? ✓

C. Will it rain in Trondheim next Tuesday?

D. What should the global discount rate be?

NUMBERS TO REMEMBER

IAMs sit behind big decisions

~1,200

what-if pathways in the IPCC's 2022 mitigation report

5

lecture days in NTNU's EP8900 course, which inspired this one

15

weeks for us to go from zero to running GCAM

IPCC = Intergovernmental Panel on Climate Change, the UN body that assesses climate research.

KEY IDEA

Who uses IAM results?

1

IPCC reports

Pathways that limit warming to 1.5 °C or 2 °C come from IAMs.

2

Governments

National climate plans test targets such as net zero by 2050 with IAMs.

3

Companies and investors

Banks and firms use IAM scenarios to stress-test their plans.

4

Researchers

To explore trade-offs between energy, food, water and climate.

WATCH OUT

Common mix-ups

Not quite	Better
"The model predicts 2.7 °C by 2100."	It says what follows if its assumptions hold. Change the assumptions and the number changes.
Process-based models tell us how much to cut.	They show how a chosen goal could be met. 'How much' is the cost-benefit question.
An IAM decides which policies governments adopt.	Policies are an input you choose. The model works out what follows from them.

Your first R script

- <- stores a value under a name
- c() combines values into a vector
- A line starting with # is a note for humans

```
install.packages("tidyverse")   # once per computer
library(tidyverse)               # every session

co2 <- c(before = 280, today = 420)   # ppm
co2["today"] - co2["before"]   # 140
co2["today"] / co2["before"]   # 1.5
```

NEW WORDS THIS WEEK

Words to add to the class glossary

Integrated assessment model A computer model that links energy, the economy, land and climate, so we can ask what-if questions about the future. Short form: IAM.

Scenario A consistent what-if story about the future, with numbers attached. It is not a forecast.

Cost-benefit model An IAM that weighs the cost of cutting emissions against the damage avoided, in a few equations. Example: DICE.

Process-based model An IAM that represents energy and land sectors in detail to show how a goal could be met. Examples: GCAM, MESSAGEix.

R package A bundle of ready-made R functions. Install it once per computer, load it with library() in every session.

HANDS-ON LAB

Lab 1: En-ROADS and your first R script

Try climate policies in the En-ROADS simulator, then install R and RStudio and run a first script.

Script: labs/lab1.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Play with En-ROADS

1

Open en-roads.climateinteractive.org. Move the sliders (carbon price, renewables, coal) and watch the 2100 temperature change. Note one question it cannot answer.

Install R and RStudio

2

Install R first, then RStudio. In RStudio make a new project called iam-course. Every lab lives in this project folder.

LAB · STEP 3 OF 4

Install and load a package

Install the tidyverse once. Load it with library() at the top of every script.

```
install.packages("tidyverse")   # once per computer
library(tidyverse)               # every session
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Run your first calculation

Open a new script, type these lines, and run them with Ctrl+Enter. Typing teaches more than pasting.

```
# CO2 in the air, parts per million
co2 <- c(before = 280, today = 420)

co2["today"] - co2["before"]   # expect 140
co2["today"] / co2["before"]   # expect 1.5
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

LAB · CHECK YOUR WORK

Did it work?

You should see

- You moved at least three En-ROADS sliders and noted the 2100 temperature
- R prints 140 and 1.5
- Your script is saved inside the iam-course project

Worksheet questions

1. Write one question En-ROADS answered and one it could not answer.
2. Add three words to the class glossary with a one-line definition each.
3. Change the script to use 2.6 ppm per year of growth: how many ppm after 10 more years?

RECAP

Take-home messages

1. An IAM links energy, economy, land and climate to answer what-if questions
2. Its results are scenarios, not forecasts: if this, then that
3. Policies are an input you choose, not something the model predicts
4. Cost-benefit models ask 'how much'; process-based models ask 'how'

Next: week 2

Week 2 opens the climate box: how CO2 becomes warming, and why the total emitted matters most.

Practise

Interactive course, week 1: try the warming explorer and the 5-question quiz.

Readings: Carbon Brief (2018), Q&A: How 'integrated assessment models' are used to study climate change; R for Data Science (free online): introduction